

ST PAUL ISLAND, NS, Canada

WMO: 714180

Lat: 47.226N

Lon: 60.140W

Elev: 27

StdP: 101.00

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.0	-12.2	-17.8	0.8	-13.9	-15.7	1.0	-12.0	27.1	-5.1	25.0	-6.1	13.4	280	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	22.5	19.1	21.4	18.5	20.5	18.0	19.9	21.6	19.2	20.8	18.5	20.1	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	21.0	18.5	13.4	20.3	17.7	12.7	19.7	57.3	21.7	54.8	20.9	52.4	20.1	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	21.2	18.9	DB	-15.6	25.3	3.4	1.2	-18.0	26.1	-20.0	26.8	-21.9	27.5	-24.4	28.3	(4)
(5)			WB	-16.1	21.1	3.7	0.5	-18.8	21.5	-21.0	21.9	-23.1	22.2	-25.8	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-3.4	-4.8	-2.7	1.0	5.3	10.4	16.5	18.2	14.5	9.5	4.6	0.1		
	DBStd	8.24	3.91	4.11	3.57	2.52	2.60	3.16	2.62	2.09	2.55	2.82	3.18	3.25		
	HDD10.0	2191	414	415	393	270	150	32	0	0	1	43	166	307		
	HDD18.3	4602	673	648	651	520	405	239	69	28	116	273	413	566		
	CDD10.0	667	0	0	0	0	3	43	200	253	137	28	2	0		
	CDD18.3	37	0	0	0	0	0	1	11	23	2	0	0	0		
	CDH23.3	14	0	0	0	0	0	0	5	9	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	9.6	12.6	10.9	10.2	9.1	8.0	7.8	7.2	7.6	9.0	10.1	10.9	12.1		
(15) Precipitation	PrecAvg	1500	148	131	117	112	101	104	88	100	119	153	153	172		
	PrecMax	1851	227	182	221	187	177	173	170	228	308	267	241	355		
	PrecMin	1151	70	48	47	59	49	51	37	44	43	61	96	97		
	PrecStd	152	38	38	42	33	36	37	30	34	60	56	40	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	6.1	7.0	11.8	16.1	21.2	23.9	24.5	22.0	18.4	14.8	10.5		
		MCWB	8.2	5.6	5.8	9.8	12.0	17.1	19.1	19.7	18.7	15.9	13.0	9.6		
	2%	DB	5.1	3.0	4.0	7.5	12.6	18.5	22.2	22.9	20.2	15.9	11.8	7.0		
		MCWB	4.5	2.4	3.2	5.8	10.1	15.0	18.7	19.3	17.8	13.9	10.7	6.4		
	5%	DB	2.9	1.2	2.1	5.3	10.4	16.6	21.0	21.8	19.1	14.4	10.2	5.6		
		MCWB	2.2	0.7	1.4	3.9	8.1	13.9	18.1	19.0	16.9	12.8	9.3	5.0		
	10%	DB	1.4	0.0	0.9	3.7	8.6	14.7	20.0	20.9	17.9	13.2	8.7	4.4		
		MCWB	0.8	-0.5	0.3	2.6	6.6	12.4	17.6	18.4	15.9	11.7	7.9	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	5.6	6.3	10.4	12.8	18.3	20.7	21.0	19.6	16.5	13.2	9.7		
		MCDB	9.5	6.0	7.5	11.3	14.2	20.4	22.6	22.9	21.2	17.9	14.4	10.5		
	2%	WB	4.8	2.6	3.5	6.3	9.9	16.0	19.7	20.2	18.4	14.3	10.8	6.5		
		MCDB	5.1	2.9	4.0	7.2	11.2	17.3	21.5	21.9	19.9	15.5	11.6	7.0		
	5%	WB	2.5	0.9	1.5	4.2	8.0	14.1	18.9	19.5	17.4	13.0	9.4	5.1		
		MCDB	2.9	1.2	2.0	5.1	9.4	15.9	20.4	21.1	18.8	14.2	10.0	5.6		
	10%	WB	0.9	-0.4	0.3	2.7	6.7	12.6	17.9	18.9	16.4	11.9	8.0	3.8		
		MCDB	1.3	-0.1	0.7	3.5	7.8	14.1	19.5	20.5	17.7	13.0	8.6	4.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.7	4.1	3.7	4.3	4.5	4.3	3.5	3.6	3.3	3.8	3.8		
		MCDBR	8.3	7.4	6.5	7.3	7.8	7.9	5.9	4.8	5.1	5.3	6.6	6.6		
	5% WB	MCWBR	8.5	7.7	6.3	6.0	5.7	5.2	3.9	3.4	4.4	4.9	6.7	7.0		
		MCWBR	8.7	7.6	6.7	7.2	7.1	7.2	5.6	4.2	4.9	5.2	6.4	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.302	0.335	0.356	0.369	0.388	0.415	0.395	0.368	0.353	0.326	0.312		
	taud		2.378	2.401	2.320	2.402	2.444	2.399	2.372	2.433	2.480	2.455	2.452	2.376		
	Ebn at Noon		767	868	885	899	896	877	846	850	844	796	747	716		
	Edh at Noon		71	85	107	109	109	115	116	105	92	81	67	64		
(44) All-Sky Solar Radiation	RadAvg		0.88	1.69	3.08	4.26	5.18	5.45	5.36	4.81	3.69	2.26	1.11	0.68		
	RadStd		0.08	0.19	0.29	0.34	0.43	0.55	0.41	0.32	0.28	0.18	0.13	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.36	N/A	N/A	N/A	N/A	N/A	N/A	+62	+12
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page